Final Project – Group Work

Amazon Reviews Analysis

Amgalan, Tseveengerel

Li, Zhongyi

Vrahliotis, Yanni

Doshi, Tanisha

Ulloa, Uriel

Bentley University

GR655-2: Managing with Analytics

16 December 2024

Executive summary

Amazon Marketplace reviews show that positive sentiment dominates with over 1.3M reviews, while neutral (300K) and negative (400K) reviews remain smaller but growing segments. However, a slight decline in positivity post-2014 suggests emerging customer dissatisfaction, while neutral sentiment is rising, indicating ambiguity on product's description leading to customers unmet expectations. Addressing these trends is critical to maintaining customer satisfaction keep growing as a prestigious company.

Key product categories reveal significant insights. Pet Products, Apparel, and Furniture receive the highest positive sentiment and should be promoted further to leverage customer satisfaction. Conversely, categories like Personal Care Appliances, Books, Beauty and Video Products show high negative sentiment, highlighting areas that require improved quality control and customer support. Additionally, factory, handling, and shipping issues are key drivers of dissatisfaction, particularly in categories like Toys, Video Products, Books, Health and Personal Care, Office Appliances and PCs, which report disproportionately high issue rates.

To address these concerns, Amazon should implement overall process improvement strategies to prioritize product quality improvements, strengthen seller accountability for categories with inconsistent experiences, address client expectations, carefully process feedback, and enhance logistics to minimize handling and shipping issues. Promoting high-performing categories while encouraging more detailed customer feedback will improve review accuracy and sentiment clarity. These actions will help Amazon sustain growth, build customer trust, and ensure a consistent, high-quality shopping experience.

Introduction

The goal of this project is to gain valuable insights for executives and decision makers at Amazon by analyzing and visualizing a data set. The findings of this data set can help drive strategic decision-making and identify areas for business improvement at Amazon. The dataset chosen, titled “Amazon US Customer Reviews Dataset,” includes insightful fields such as product Category, rating, review headline, review body, review date. Hidden nuances can be found to better understand the data at hand which can provide Amazon with insights to improve business processes. Some of the insights that were derived from the data set in this project include sentiment analysis and customer satisfaction trends, rating patterns and product attributes, customer behavior patterns, and text mining for product improvement insights.

Description of the Database

The dataset includes key variables such as comment metadata, sentiment analysis metrics, and issue identification fields. Product Title Indicates the name of the reviewed product. The Star Rating field provides a rating from 1 to 5 stars which reflects customer satisfaction. Product categories divide products into broad categories. Total Votes indicates the number of useful votes a comment received. For sentiment analysis, the dataset includes variables such as the compound score, which is an aggregate sentiment score from -1 to 1 that represents the overall mood of the comments. Other fields include Positive, Neutral, and Negative Scores, which provide specific measures of emotional polarity. The "Sentiment" field classifies each comment as "positive," "neutral," or "negative," providing a direct summary of the sentiment.

The dataset also provides detailed issue identification fields that highlight specific issues mentioned in the comments. Factory problem flag review, involving product defects or quality issues. Shipping problem determination discusses the review of delivery delays or errors. Refund

and Return issues highlight comments related to refund or return challenges. Processing issues focus on comments about damaged or mishandled products.

Another aspect of this dataset is the relationship between key review metrics, including product categories, star ratings, help votes, and total votes, which could help us to find the relationship between the number of reviews, the rating liability, and polarizing product category.

We used this dataset to identify polarized products. Using the variability and range of star ratings to identify products where opinions differ while highlighting categories where products are consistently polarizing. Customer behavior analysis is another area that we investigated, where we examined buying relationships across categories and identified products or categories that are prone to pain point triggers.

Understanding and analyzing customer behavior is also important. Customer satisfaction trends can be explored by examining which product categories have the highest and lowest level of customer satisfaction. Finally, text mining techniques can be applied to data sets to uncover common consumer pain points and identify patterns of dissatisfaction across product categories. These insights can improve product quality and enhance the customer experience.

This dataset provides a comprehensive basis for analyzing customer behavior, satisfaction trends, and product-related issues. It provides deep insights to improve product quality, customer experience, and identify areas that need to improve.

Methodology:

1. Understanding the Dataset Explore the dataset to identify its structure and key features, such as review body for sentiment analysis. Check for missing values and assess the dataset's relevance to the analysis objective of evaluating customer sentiment.

2. Data Preparation Clean the data by handling missing values and removing irrelevant columns. Sample the dataset for efficiency if needed and label the data manually with external sentiment classifiers where necessary.

3. Text Preprocessing Prepare text data by tokenizing, lowercasing, and removing stopwords, punctuation, and special characters. Apply lemmatization to reduce words to their base forms for consistency and clarity. Categorize each token as a part of speech to extract the adjectives used in each review.

4. Exploratory Data Analysis: Visualize sentiment distribution and common words using plots or word clouds. Analyze the relationship between review lengths and sentiment to uncover patterns in customer feedback.

5. Feature Engineering Transform text into numerical data using techniques available in Natural Language Processing packages. These methods capture textual relevance and context for machine learning models.

6. Insights and Visualization Analyze trends in sentiment across time or product categories. Use topic modeling to uncover key themes in reviews and derive actionable insights for improving customer satisfaction.

This methodology provides a concise and structured approach to understanding and analyzing customer sentiment effectively.

Results

Sentiment Analysis and Customer Satisfaction trends

Sentiment Overall Analysis

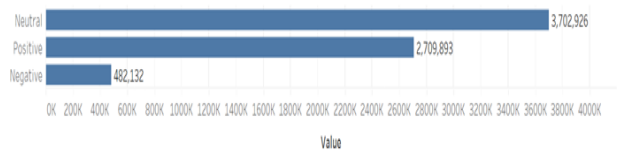


Figure 1 Total Number of Sentiment Score

The distribution of sentiment scores classified as positive, neutral, and negative is around 6.9 million. Neutral sentiment leads with 3.7 million occurrences, 53.7 %, followed by positive sentiment at 2.7 million occurrences, 39.3 %. The negative sentiment score is lower, around 0.5 million, at around 7 %. The distribution implies that customer reviews are neutral and positive, revealing common satisfaction.

Sentiment Average Analysis

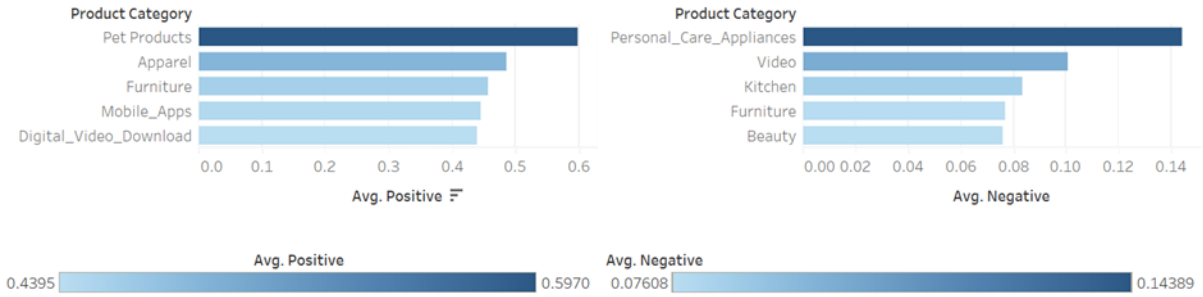


Figure 2 Top 5 product category by positive and negative sentiment

The analysis of positive and negative sentiment across product categories explores shifting levels of customer satisfaction. Amazon customers are positively satisfied with pet products, with an average positive sentiment of 0.60, followed by Apparel, Furniture, Mobile Apps, and Digital Video Downloads. On the other hand, customers are negatively satisfied with Personal care Appliances, average is 0.14, followed by Video, Kitchen, Furniture, and Beauty, implying problems may need consideration. Amazon may need to pay more attention to customer care, product quality, and delivery in those negative sentiment product categories.

Sentiment Proportional Analysis

This segment provides an analysis of the specific words used on reviews based on proportional values (negative and positives reviews over total reviews) to clearly analyze each category impact.

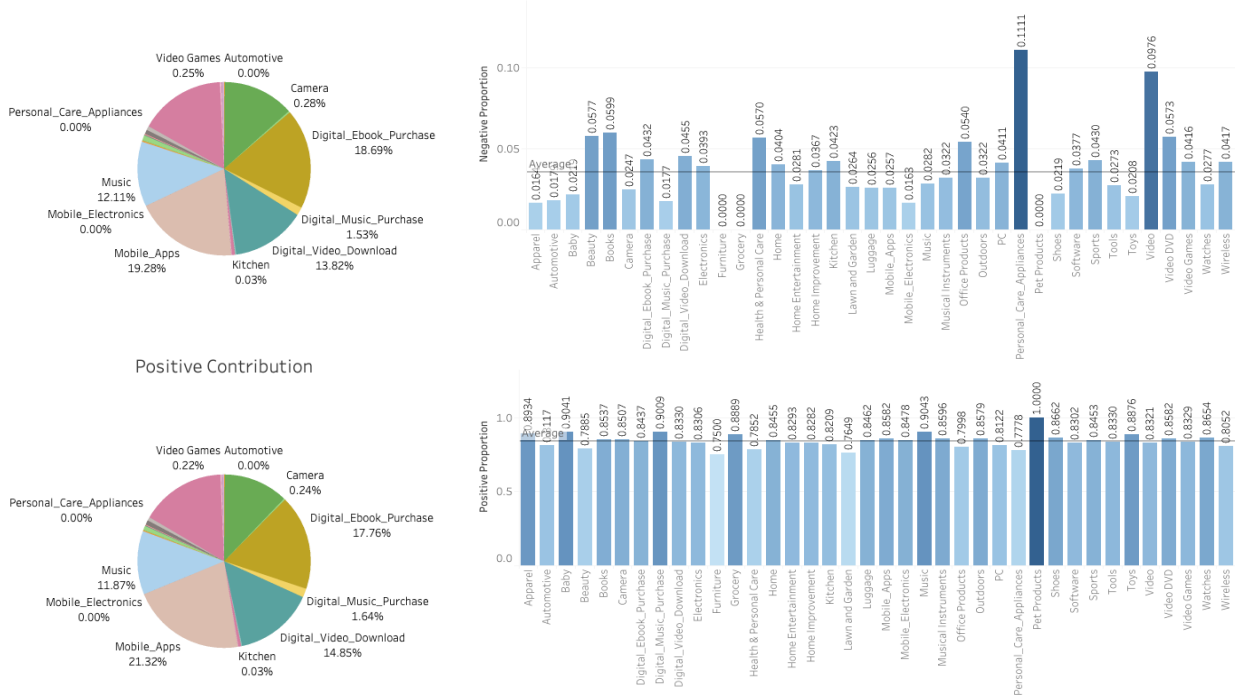


Figure 3 Sentiment Contribution vs Sentiment Proportion

The graph provides a detailed breakdown of positive and negative contributions by product categories based on customer reviews. It includes pie charts showing proportional contributions of each category to positive and negative feedback and bar charts comparing the proportions across all categories.

On one hand, categories like Apparel, Grocery, Baby and Pet Product maintain consistently high positive review proportions (around 80%), reflecting strong customer satisfaction. Conversely, outliers like Personal Care Appliances, which show a disproportionate spike in negative proportions (11%), required deeper investigation to uncover specific challenges. The graph identifies Beauty, Books, Health & Personal Care, Office Products, Personal Care Appliances, and Video as the categories with the highest negative review proportions (above the

standard/average 3%), requiring targeted improvements. These categories likely face challenges related to product quality, usability, or mismatches between customer expectations and product delivery.

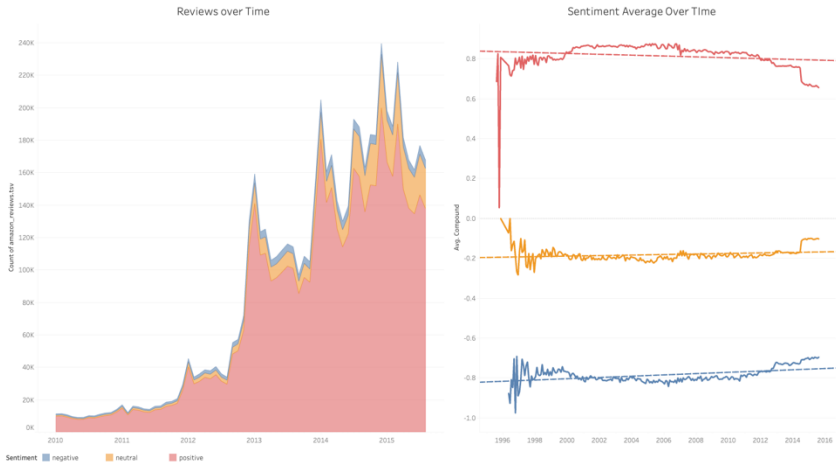
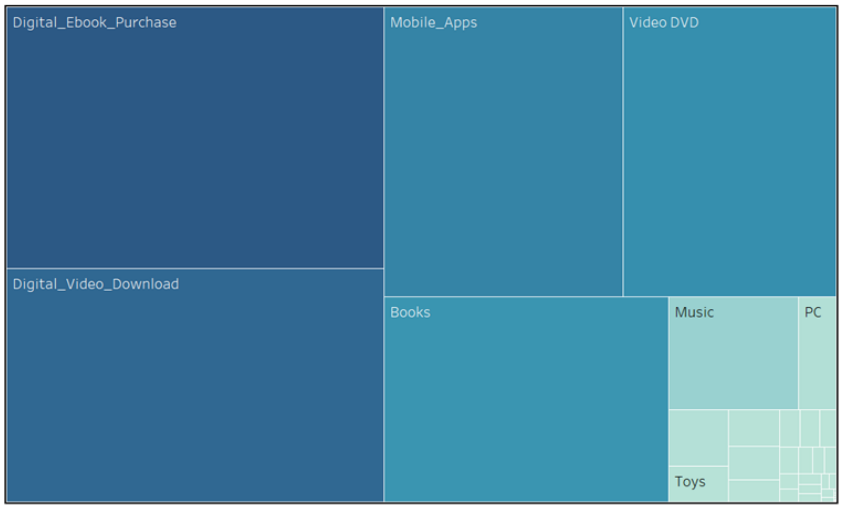


Figure 4 Review by sentiment period

The graph suggests there has been a significant grow of the total number of reviews from 2010 to 2015, with noticeable peaks around 2013 and 2014, which reflects an increasing customer engagement that might be the result from higher marketplace and product offering.

Positive reviews dominate the overall time, meanwhile negative and neutral ones remains as the smallest part, having a constant proportion over time. Regarding the average compound (this value determines whether a visit is positive, >0.5, neutral, in between -0.05 and 0.5, and negative, less than -0.5). Positive reviews remains high (~0.8) but shows a slight downward trend starting around 2014. This could indicate emerging dissatisfaction in specific areas. Neutral sentiment has remained relatively stable but slightly upward since 2014, suggesting customers increasingly express non-committal feedback. Negative sentiment has seen a slight improvement (less negative compound scores) since 2012, but it remains consistently lower than positive scores. The slow upward trend indicates that while issues persist, Amazon may have taken steps to address common complaints, improving consumer satisfaction, extending valuable insights for sustaining or reproducing these results in the future.

Pain Point Analysis



Product Category. Color shows count of Issues.csv. Size shows count of Issues.csv. The marks are labeled by Product Category. The data is filtered on Running Sum of Count of Issues.csv and Review Date Year. The Running Sum of Count of Issues.csv filter ranges from 0 to 62,864. The Review Date Year filter keeps 6 of 21 members.

Figure 6 Total of issues per product category

By understanding which product categories experience the most issues at Amazon, inventory managers can predict which product categories will experience the most returns and refunds. This can help inventory managers decide how much inventory of a product they should carry as product categories with higher return rates likely do not need as much inventory. This can lead to more efficient use of warehouse space and help avoid overstocking certain products. Figure 4 shows that video downloads, video DVDs, eBooks, and books had the most issues out of any product category at Amazon, meaning managers can anticipate the most returns from these product categories.

Factory Issue	Shipping Issue	Handling Issue No	Handling Issue Yes
No	No	200,642	13,496
	Yes	34,556	20,073
Yes	No	11,227	1,908
	Yes	4,563	3,886

Count of Issues.csv broken down by Handling Issue vs. Factory Issue and Shipping Issue. Color shows count of Issues.csv. The marks are labeled by count of Issues.csv.

Figure 5 Issues product had per category

Decision makers can also determine what the issue was with the product to determine what can be done in the future to limit issues. If the issue is a factory or quality issue, then Amazon can look at which suppliers are producing those products with issues and possibly use products from vendors with higher quality products. If the issues are from handling or shipping, then Amazon can look to make improvements in those areas. As shown in figure 5, most issues from Amazon products were not from shipping, handling, or factory issues, and are instead refund and

return issues which can be improved by changing refund and return policies. The most common non refund and return issue was shipping issues at 34,556 issues, while the least common was factory issues at 11,227 issues.

Consumer Pain Points Proportion per Category Analysis

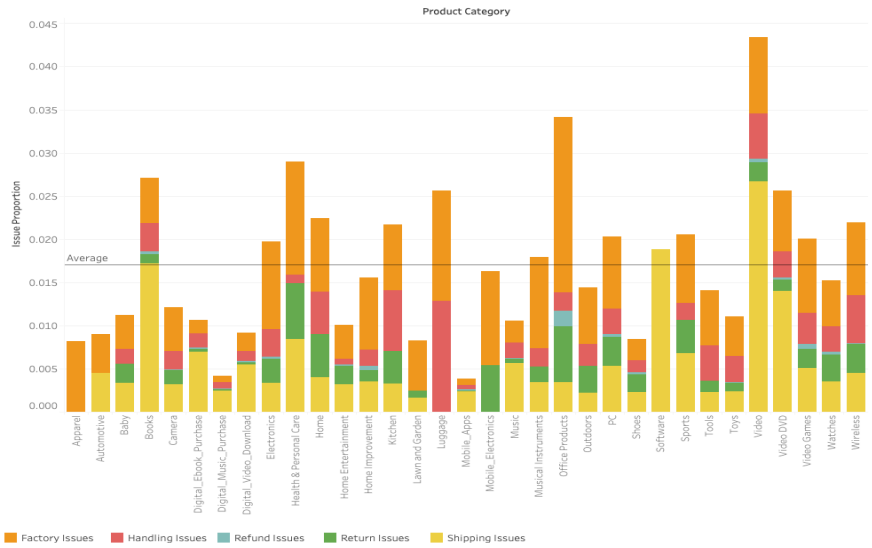


Figure 7 Pain Point Proportion

The stacked bar chart represents the proportion of issues expressed by customers in their reviews based on the total amount of reviews per category. Note this analysis was computed based on the proportion and not on the total amount of issues since there is a large gap between the top reviewed categories and the bottom ones, which would not allow to properly analyze each issue/category.

The graph highlights significant variations in customer issues across categories, note there is an average of about 1.7% issues presented per category. Departments such as Video, Office Products, Healthcare & Personal Care and Books far exceed the average. The most common problems presented were Factory Issues (poor quality or defects) and Shipping Issues (Delayed or damaged deliveries), indicating challenges in quality control and logistics. On the other hand, categories like Digital Video, Mobile Apps, and Digital Video have consistently low issue proportions due to their nature.

A deeper look at issue types reveals that Factory Issues dominate in most categories, particularly in Apparel, Office Products, Health & Personal Care and Luggage, suggesting a need

for enhanced manufacturing standards and pre-shipment quality checks. Likewise, shipping issues represents a notable problem for Books and Video related products. Additionally, Handling Issues are prominent in Kitchen and Luggage categories. Lastly although in a less frequent proportion Refund and Return Issues are prevalent in Mobile Electronics, Health & Personal Care and Office Products.

Rating Analysis

To investigate the rating patterns and product attributes within Amazon, we conducted an analysis that explores the relationship between the number of reviews and rating variability across different product categories providing insights into how review volume impacts the reliability of customer ratings.

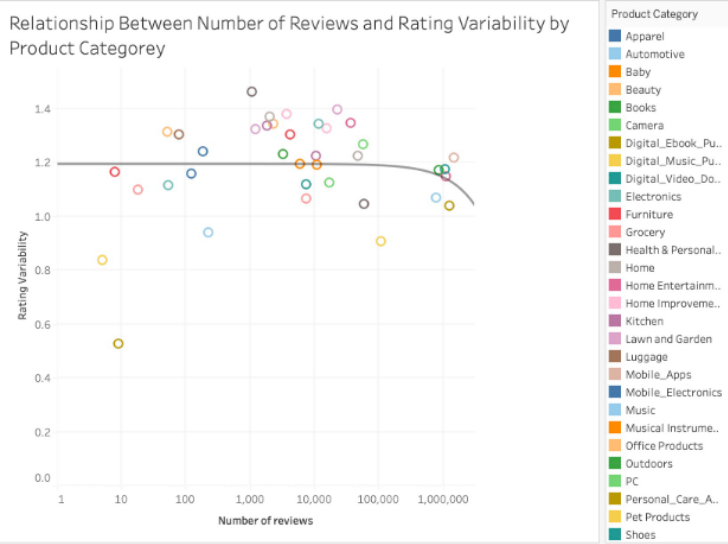


Figure 8 Relationship Between Number of Reviews and Rating Variability by Product Category

Figure 8 - Relationship Between Number of Reviews and Rating Variability by Product Category

The scatter plot illustrates a negative correlation between the number of reviews and rating variability, according to Figure 6. Variability is higher for products with fewer than 10,000 reviews, from 1.0 to 1.4, due to the disproportionate impact of individual ratings. However, products with more than 10,000 reviews exhibit reduced variability, usually below 1.0, reflecting stabilized ratings as the influence of outliers diminishes. For example, ratings for categories such as "books," "electronics," and "home improvement" stabilized faster and closely correlated with trend lines. In contrast, the "baby" and "beauty" categories showed higher variability in lower review counts, possibly due to

inconsistent product quality or different customer expectations. These indicate the importance of encouraging more reviews to reduce variability and improve rating reliability.

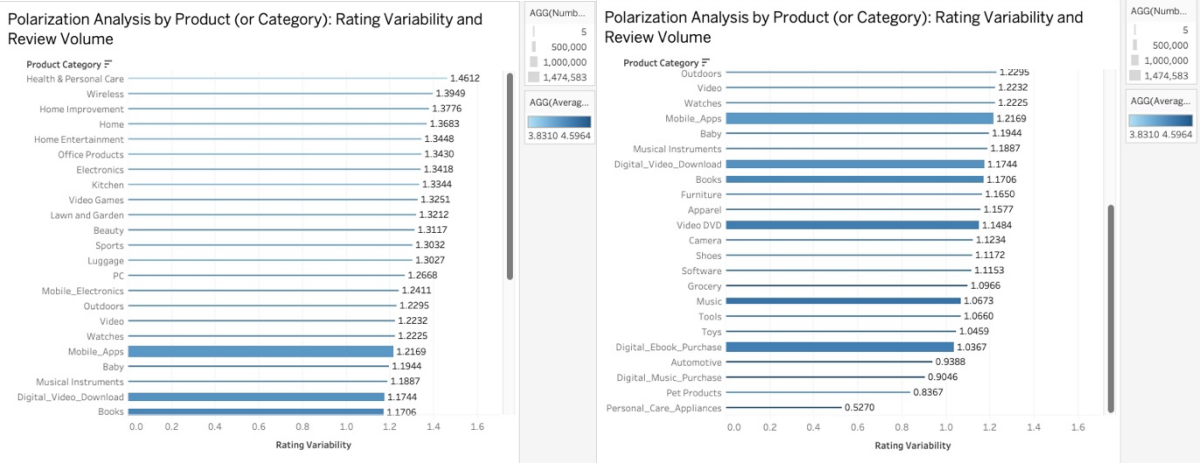


Figure 9 Polarization Analysis

Figures 9 are bar charts that show the differences in rating variability across product categories. Rating variability is measured by the standard deviation of star ratings, which identifies categories where customer experience is highly inconsistent. These charts focused on findings related to the high and low variability categories and their implications.

The most polarized category is health and personal care, with a rating variability of 1.4612. Wireless and Home Improvement also show high variation, which also shows a polarizing experience. For these categories, the broad range of products and their use cases may facilitate comparative review as customers have different needs and standards. In contrast, Personal Care Appliance and Pet Products display the lowest variability, indicating more consistent customer satisfaction. These categories may feature niche products that align well with customer expectations or have fewer quality control issues, leading to more uniform positive ratings.

Polarizing categories such as health and personal care, wireless, and home improvement reveal areas where customer experiences vary widely. Addressing variability in these categories may involve improving product quality control or aligning product marketing more closely with

actual customer expectations. At the same time, categories with lower variability, such as personal care devices, provide insights that maintain consistent customer satisfaction. These findings can help businesses identify opportunities, refine their offerings, and increase customer trust.

Text Mining and Product Improvement Insights

In this section insights on customers preferences over reviews would be detailed to provide further analysis on which are the most client-sensitive features of the categories with highest number of negative comments and address them to provide recommendations.

Figure 10 Sports Wordcloud

For Sports & Outdoor Equipment, terms like knife, sheathe, blade, pulse, and steps, suggest demand for durable survival tools and wearable fitness trackers. Satisfaction is high, but there's room to enhance product durability, introduce multi-functional tools, and improve wearable tech features like app integration and battery life. Bundling related products can also boost value for customers.

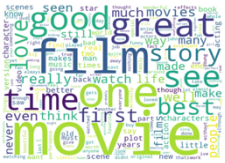


Figure 11 Video WordCloud

Video consumers mostly interact with series and episode-like content, particularly in the action-genre and emotionally resonant content. There is a high demand of quality narratives and captivating shows. In

this regard, Amazon could enhance Prime Video by expanding original content, improving personalization and integrating user feedback.

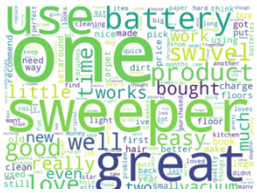


Figure 12 Home WordCloud

The most important factors in home products as per user's review are ease of use, reliability, and battery performance, being sweepers the most reviewed cleaning product. Features such as simple design and effective functionality are highly appreciated, and battery life remains the greatest concern. Amazon can promote eco-friendly cleaning solutions with bundled offerings for home essentials.

In Personal Care, key terms such as hearing, pressure, device, and easy, reflect a high demand for health-monitoring and personal care technologies. Customers value ease of use and functionality. Amazon should focus on improving usability, expanding health tech offerings like hearing aids and blood pressure monitors, and integrating customer feedback to refine features like battery life and app connectivity.

Figure 13 Personal Care

Figure 14 Book WordCloud

Consumers appreciates engaging stories and memorable characters, with a strong preference for series. Quality storytelling is highly valuable due to the love and favorite terms highlighting the importance of personal connections. There is a chance of leveraging engagement by promoting this kind of book bundles and enhancing algorithm recommendations.

Electronics demands sound quality, ease of use and value for money, in this regard headphones are the key product focus. Highlighting features like noise cancellation and offering budget-friendly options under Amazon Basics can capture price-sensitive consumers. Additionally,

Figure 15 Electronic WordCloud

offering audio products with Amazon Music subscription could prove to be a strong bundle for growth opportunity.

Figure 16 Beauty WordCloud

The Beauty category highlights a strong overlap with entertainment, as terms like movie, dvd suggest demand for themed beauty products inspired by popular franchises. Positive words like great, and first, indicate initial satisfaction with product quality. Amazon can leverage this by launching collaborations with media franchises, emphasizing unique branding, and encouraging detailed reviews to better understand customer.

Recommendations

Based on figure 5, Number of Issues by Product Category, we can recommend that Amazon update their models that help forecast returns based on the product categories that have the most and least number of issues. This will help managers improve inventory efficiency and avoid overstocking certain product categories.

Amazon must prioritize customer issues by product category and leverage main insights to resolve product quality, return delivery, and customer care. Based on seasonal differences, Amazon should enhance targeted initiatives during peak periods and sustain increased customer satisfaction.

Products with fewer reviews tend to show greater variability. To stabilize ratings and improve their reliability, businesses can encourage more customer reviews by offering incentives such as loyalty points, discounts, or follow-up reminders post-purchase.

To continue with the upward trend of reducing the impact of negative comments, Amazon should develop the following strategies:

Customer Centric retention: Create a smooth channel that would rapidly address the concerns of dissatisfied customers through faster dispute resolution and product guarantee.

Product transparency: Improve accuracy of product descriptions, sellers performance and delivery timelines.

Feedback Loop: Provide actionable data to sellers to help them improve customer satisfaction and reduce negative reviews.

Customer engagement: Incentivized customers to write reviews by introducing enticing rewards for verified purchases.

Conclusions

The Amazon customer reviews study explores strengths and areas for improvement across products and product categories. The Pet Products, Apparel, Grocery, and Baby categories attained steadily high positive sentiment scores, while the Health and Personal Care, Personal Appliances, Beauty, and Video categories showed higher negative sentiment score proportions, requiring targeted interventions. The most frequent complaints from Amazon consumers involved refunds and returns, followed by shipping and factory defects. Customers of the Video, Books, and Office Products categories experienced the highest rates of criticism, highlighting the need to focus on quality control and strengthen consumer care.

Amazon's Overall sentiment trends are positive, reflecting strong customer satisfaction. However, there is still room for improvement, emerging neutrality and slight downward positivity since 2014 indicate opportunities to grow. By addressing root causes of negative reviews and promoting transparency and quality will further enhance customer trust and loyalty. Based on these insights, Amazon should rank solving category-specific challenges. It also must consider which products are dominated by negative sentiment, then change the return policy, improve product quality, and use innovative AI for customer feedback.

References

Varasteh, A. (2024). *Python NLTK for Beginners: Customer Satisfaction Analysis* [Guided Project]. Coursera.

https://www.coursera.org/projects/python-nltk-for-beginners-customer-satisfaction-analysis

Appendix

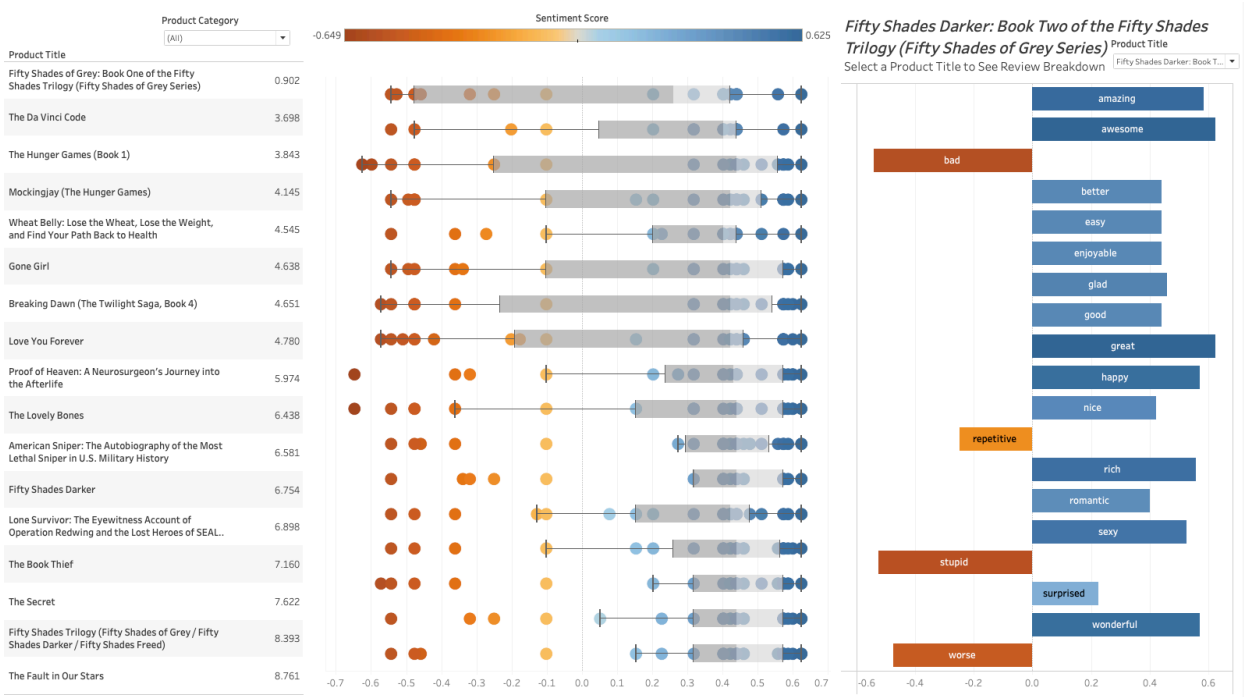


Figure 17Video Sentiment Analysis Product Performance

The previous graph represents a sentiment analysis breakdown by product title, this feature would help to drill deeper into each product category and find out which product title are the ones contributing the most to the negative comments. Due to the project scope this analysis won't be further included.